

Refining wind and solar potential maps through spatial multicriteria assessment. Case study: Colombia

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ABSTRACT

Despite the abundance of energy resource assessments for wind and solar power around the globe, the existing studies seldom integrate the environmental restrictions into the analysis, which results in overestimation of the available resources. Existing potential maps show that wind and solar resources are abundant in Colombia, but these studies do not consider the multiple restrictions developers face when a project is considered. This paper proposes a methodology to determine the areas with the best characteristics to develop wind and photovoltaic solar farms in Colombia, using Geographic Information Systems that depict the distribution of the physical, biotic, economic, cultural, and political characteristics restricting or conditioning the implementation of these projects. To consider financial performance, the LCOE associated with each location was subsequently calculated, and the environmental characteristics were assessed through interviews with a group of experts. Maps were developed where the areas with potential can be viewed, classified according to their profitability and the ease of developing a project at that location. The most promising locations for wind and solar farms for the continental territory of Colombia were determined integrating levels of difficulty in the environmental management, and financial project performance. The results were unified in a tool useful for stakeholders, decision-makers, researchers, and investors, in which basically no technical knowledge is required for its interpretation. The results allow estimating the real installable potential in each technology throughout the Colombian territory and providing signals for the national interconnected system (SIN) on where the country's new energy developments are expected to take place.

Introduction

The incorporation of energy generated from non-conventional renewable sources has been growing on a global level in recent years, in order to comply with the different agreements reached in international conventions and meetings (BP, 2020). The abundance of any resource is not the only necessary condition for their exploitation to be feasible. Studies have been carried out in several countries to determine the areas where the greatest solar or wind energy potential is found (Ali et al., 2019; Ángel-Sanint et al., 2010; Anwarzai & Nagasaka, 2017; Aydin et al., 2010; Baban & Parry, 2001; Brewer et al., 2015; Cerón & Viatcheslav, 2015; Delicado et al., 2016; Doorga et al., 2019; Giamalaki & Tsoutsos, 2019; IDEAM, 2015a,b; Janke, 2010; Latinopoulos & Kechagia, 2015; Mentis et al., 2015; Omitaomu et al., 2012; Pinilla et al., 2009; Rodríguez-Urrego & Rodríguez-Urrego, 2018; Sánchez-Lozano et al., 2016; Siyal et al., 2015; Teske et al., 2019; UPME, 2018; UPME &

BID, 2015; Van Haaren & Fthenakis, 2011; Watson & Hudson, 2015; Yang et al., 2019; Yushchenko et al., 2018), but few of them integrate the environmental restrictions that would prevent developing this type of project, thus being able to move from estimating the theoretical or technical potential to the sustainable potential (Teske et al., 2019). This lack of analysis overestimates the usable potential of renewable energy sources, and at the same time fails to address local conditions that may complicate or impede project development. The present paper builds upon the existing knowledge not by only proposing a methodology that identifies the areas where project development is restricted by different factors, but also by identifying the most promising locations by jointly analyzing two factors: the level of difficulty in the environmental management and the financial performance of the projects.

This scope can be used to significantly reduce the costs to be assumed in environmental management (Ángel-Sanint et al., 2010), diminishing the uncertainty of expected returns on investments (Delicado et al.,

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2016) and lessening the risks of unforeseen conditions that may delay project development or its feasibility itself. Responsible planning for electricity generation projects from non-conventional renewable sources will foster the environment of acceptance and legitimacy from stakeholders, for these types of projects to be implemented as smoothly and quickly as possible (Cerón & Viatcheslav, 2015).

Colombia is a country with abundant renewable energy resources. The most attractive wind resource is located the Caribbean region, especially La Guajira (IDEAM, 2015a), the department hosting the only wind farm in Colombia since 2004 (Pinilla et al., 2009), where most projects have been registered for this kind of technology. Solar radiation is abundant throughout the territory, with a country average that is higher than the global average. However, regions such as the Pacific and the Amazon have radiation values that are not attractive for developing solar PV energy projects on commercial scales (IDEAM, 2015b; Rodríguez-Urrego & Rodríguez-Urrego, 2018). The Institute of Hydrology, Meteorology and Environmental Studies (IDEAM) has undertaken the task of estimating the potential of resources and characterizing them in Colombia (UPME & BID, 2015), but this approach has been merely from the perspective of the physical magnitude of the resource, and neglects other project requirements.

The installed capacity of the Colombian energy sector is mainly composed of hydroelectricity (65 %) (UPME, 2018), and the remaining 35 % comes from thermoelectric plants that burn different fossil fuels. However, in a normal year, up to 85 % percent of the energy is generated using hydro, and most fossil capacity is used to ensure reliability. This situation gives Colombia a comparatively low carbon-intensive electric system. At the same time, it makes it susceptible to extreme climatic events, especially droughts during the warm phase of the El Niño Southern Oscillation - ENSO (Henao & Dynner, 2020).

To diversify large-scale generation sources in Colombia, Law 1715 of 2014 was sanctioned with the objective of incorporating non-conventional renewable sources, reducing emissions from the energy sector and promoting the adoption of renewable energy by end-users (Congreso de la República de Colombia, 2014). In the renewable energy auction carried out in 2019, 1374 MW of solar and wind energy were assigned to start generating in 2022 (although there have been important delays in their development). It is expected that, as time goes on, participation in renewables will end up accumulating >5000 MW of installed capacity (Presidencia de la República de Colombia, 2023).

Renewable resource assessment has been carried out in Colombia, focused mainly on the Guajira or the Caribbean region in the north of the country (Carvajal-Romo et al., 2019; Pabón-Hernández, 2017), integrating, at most, some environmental conditions that would restrict the implementation of this sort of project. Despite this, to date there is no concrete value for the installable potential of solar and wind energy on commercial scales in Colombia, and there is no public roadmap to evaluate the areas with the greatest feasible potential. A clear understanding of the local environmental characteristics could foster more effective project development processes, as it allows quantifying the feasible potential in the Colombian territory for each of these technologies with less uncertainty.

This research aims to jointly assess the spatial distribution of renewable energy resources (wind and solar PV) with the environmental restrictions and criticalities that condition implementing this type of projects in the Colombian continental territory. This allows to estimate the existing potential for each of these sources, to properly identify the feasible regions on the country for project development and subsequently to it according to the greater or lesser financial performance and the degree of effort in environmental management that it entails.

Materials and methods

The methodology is based on a spatial multi-criteria decision-making process, made up of an independent analysis of environmental conditions that may affect the implementation of solar and wind parks. It also

includes a financial performance assessment of these power plants, which will be integrated afterwards to determine the most promising locations for deploying these non-conventional renewable energy projects.

The environmental component of the work is mainly based on the methodology proposed by Ángel-Sanint et al. (2010), that set the basis for assessing the environment through an analytical method by dimensions. Moreover, we include elements from Teske et al. (2019), where the authors propose a definition of sustainable potential that includes different ecological and socioeconomical aspects, which significantly reduces the potential assessed considering only the resource and the technology.

These investigations make it possible to focus research, studies, or developments in the areas where favorable technical and socioeconomic aspects converge. This paper considers these experiences to propose a methodology that includes both the environmental restrictions and the performance of the projects.

The multicriteria includes different environmental dimensions of analysis: physical, biotic, economic, cultural and political. It is worth noticing that the economical dimension is considered in two different ways in this paper. The first considers the economic impacts of any particular project in the multicriteria analysis, while the second uses financial tools such as a distributed calculation of the Levelized Costs of Energy (LCOE) in the Colombian territory, to assess the attractiveness of developing a project. A diagram of the methodology is presented in Fig. 1. Seven steps are involved, as described below:

Step 1: environmental restrictions

A systemic review allowed identifying the environmental characteristics that restrict the development of a wind farm or photovoltaic solar farm. However, Colombia offers some features that require specific assessment. The restrictions were identified considering the 5 environmental dimensions defined by Ángel-Sanint et al. (2010), these are defined as follows:

1. Physical: Impacts on soil, water, and air
2. Biotic: Impacts on fauna, flora, ecology, and ecosystem integrity
3. Economic: Impacts on the production and trade of goods and services
4. Cultural: Impacts regarding the adaptation of a particular human group to its surrounding environment
5. Political: Impact on the existing power structure and the potential conflicts

Some of these restrictions were created by law to protect ethnic groups, ecosystems, economic activities, and others, but they can also be due to activities or actions related to the country's internal armed conflict. Table 1 shows the environmental restrictions defined for both Solar PV and Wind Farms.

Step 2: environmental criticalities

In areas without restrictive characteristics that make it impossible to develop wind or solar power generation plants, there may be conditions that add complexity to the environmental management necessary to ensure the project's feasibility in that location. They may be caused by land cover, land use, social or cultural conditions or the ways existing populations or individuals in the biotic environment relate to them. In turn, these characteristics can determine the areas that have an attractive potential for developers and are more suitable due to the relative ease they offer to carry out the necessary processes in any phase of the project (UPME et al., 2017). Fig. 2 shows different characteristics that were considered as criticalities and dimension to which they belong.

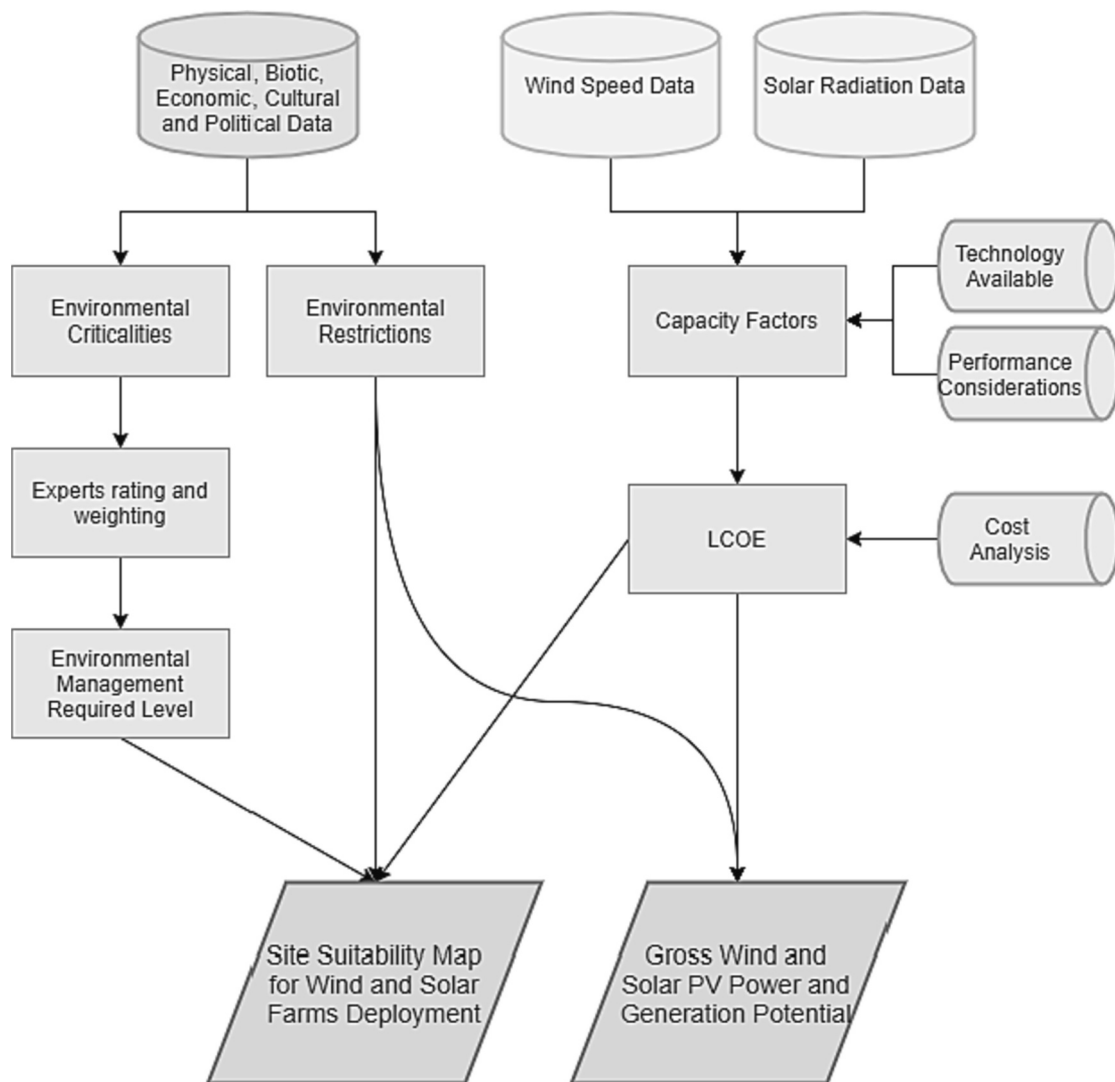


Fig. 1. Graphical scheme of the methodology developed

Step 3: preference structure

A group of experts representing different institutions and economic sectors were surveyed to assign qualifications to the characteristics shown in Fig. 2 and make it possible to integrate them in numerical values that allow further evaluations. The experts were selected to match the following criteria:

- All of them have outstanding knowledge in at least one of the dimensions in which the environment was assessed
- The experts belong to public and private companies and sectors such as education, environmental licensing, project development, and consulting.
- At least 10 years of experience in the Colombian electric and/or environmental sector

The experts were consulted in a semi-structured interview. The experts participated voluntarily and were duly informed about the scope and objectives of the present work. They were asked to respond based on their personal professional criteria, noting that their answers would not represent the opinions or criteria of the institutions they work with. The experts explicitly allowed the researchers to use their responses in the present work. The sector and field of expertise of the surveyed experts are shown in Table 2.

Individual guided surveys were held with each expert to rate the criticalness of the different environmental characteristics, in order to integrate them into a consolidated result. A score between 0 and 10 was asked to be provided for any independent characteristic, where 0 meant the lowest environmental criticality and 10 denotes that it implied dedicated and expensive environmental management to overcome that condition. Results are shown in Table 3.

Experts were also asked to distribute 100 points among the environmental dimensions (Physical, Biotic, Economic, Cultural) to find their relative weight. Distributing the points grants finer answers because giving more points to one item forced taking them from another. This procedure helps reduce bias in determining the preference structure (Kumar et al., 2017). Results are shown in Table 4. It is worth noticing that the political dimension was not assessed due to the absence of characteristics considered as criticalities.

Step 4: required level of environmental management

The results obtained in 2.3 are the input values for different layers that were integrated through ArcGIS 10.4.1 using ArcMap, ArcCatalog and their Model Builder Tool. The results are presented as maps, where the spatial distributions of criticalities and their management complexity are displayed.

Table 1
Environmental restrictions for solar PV and wind farms.

Variable	Dimension	Project	Condition	Reference
Energy potential	Physical	Solar/ wind	Non profitable	(Baban & Parry, 2001; Cevallos-Sierra & Ramos-Martin, 2018; Höfer et al., 2016; Rodman & Meentemeyer, 2006; Tegou et al., 2010; Van Haaren & Fthenakis, 2011)
Waterbodies	Physical	Solar/ wind	<60 m/<165 m	(Baban & Parry, 2001; Höfer et al., 2016; Tegou et al., 2010)
Wetlands	Physical	Solar/ wind	<60 m/<165 m	(Cevallos-Sierra & Ramos-Martin, 2018; Höfer et al., 2016; Rodman & Meentemeyer, 2006; Tegou et al., 2010)
Surface roughness	Physical	Wind	Forest	
Flood risk	Physical	Solar	Flood zones	
Airport	Physical	Wind	<5500 m	(Tegou et al., 2010)
Slope	Physical	Solar/ wind	>10 %/>20 %	(Baban & Parry, 2001; Cevallos-Sierra & Ramos-Martin, 2018; Höfer et al., 2016; Rodman & Meentemeyer, 2006; Tegou et al., 2010; Van Haaren & Fthenakis, 2011)
Available area	Physical	Solar/ wind	<100 ha/<9 km ²	
Protected areas	Biotic	Solar/ wind	National System of Protected Areas (SINAP)*	(Baban & Parry, 2001; Cevallos-Sierra & Ramos-Martin, 2018; Höfer et al., 2016; Rodman & Meentemeyer, 2006; Tegou et al., 2010)
High ecosystemic value zones	Biotic	Solar/ wind	Páramos, mangroves, bogs	(Cevallos-Sierra & Ramos-Martin, 2018; Höfer et al., 2016; Rodman & Meentemeyer, 2006; Tegou et al., 2010)
Birds habitats	Biotic	Wind	Halobiomes y Hellobiomes	(Höfer et al., 2016; Van Haaren & Fthenakis, 2011)
Visual impact and urban expansion	Economic	Solar/ wind	500 m < 2000 m (populated areas)	(Baban & Parry, 2001; Cevallos-Sierra & Ramos-Martin, 2018; Höfer et al., 2016; Rodman & Meentemeyer, 2006; Tegou et al., 2010; Van Haaren & Fthenakis, 2011)
Distance to roads	Economic	Solar/ wind	30, 45, 60 m/<165 m	(Baban & Parry, 2001; Cevallos-Sierra & Ramos-Martin, 2018; Höfer et al., 2016; Tegou et al., 2010; Van Haaren & Fthenakis, 2011)
Distance to electric grid	Economic	Solar/ wind	60 m/<165 m	(Baban & Parry, 2001; Cevallos-Sierra & Ramos-Martin, 2018; Höfer et al., 2016; Tegou et al., 2010; Van Haaren & Fthenakis, 2011)
High archeological value areas	Cultural	Solar/ wind	Protected archeological areas	(Baban & Parry, 2001; Höfer et al., 2016; Tegou et al., 2010)
Ethnic territories	Cultural	Solar/ wind	Ethnic groups reservations	
ETCR - Territorial Spaces for Capacitation and Reincorporation	Political	Solar/ wind	Territorial Spaces for Capacitation and Reincorporation (ETCR)	
Coca crops	Political	Solar/ wind	Presence of coca crops 2018–2019	

Step 5: energy generation

The amount of energy generated by the plant is one of the main inputs to calculate the LCOE. To estimate the potential energy generation in a country-wide scale, the resource was multiplied by the capacity factors (CF) for wind and solar plants.

For 100 MW Solar PV parks with fixed structures, the annual mean of Global Horizontal Irradiation (GHI), taken from the Solar Atlas developed by EIA University (González-Duque et al., 2018), is converted to energy generation. This data is later affected by the general losses due to ambient temperature, fixed orientation and tilt, module technology and availability of the plant, as suggested in Resolution 201 of 2017 of the Gas and Energy Regulation Commission of Colombia (CREG) (CREG, 2017).

For 100 MW wind farms, hourly wind speed data, derived from the ERA5 Land project of the European Centre for Medium-Range Weather Forecasts - ECMWF (de Assis Tavares et al., 2020), needs to be extrapolated to the height at which wind turbines operate. Then, the power curve of a specific generator must be used to determine the power output and energy generation during a given period. To achieve better results, the work performed by Ángel-Sanint et al. (2020) was implemented, assigning generic characteristics of wind turbines depending on the IEC Wind Classification. This assignment was based on the evaluation of commercially available technology. Afterwards, the park's performance was affected by layout losses, availability factors and common losses for these types of power plants (Pabón-Hernández, 2017). As there is no linear relationship between wind speed and power output, higher time resolution would lead to less errors in calculations.

Step 6: levelized cost of electricity (LCOE)

The LCOE will be calculated for each technology as a spatial variable within the entire Colombian territory to integrate the financial behavior of the projects within the proposed multi-criteria objective analysis. This metric considers the costs of the project, both investment and O&M, and the amount of energy generated. Therefore, it integrates the performance of the project, investments necessary to achieve it and expected profitability by the owners of the capital. Thus, it is taken as a profitability index associated with each project at any assessed location.

The LCOE is widely used by authors and institutions when trying to compare the unit cost of energy between various proposed generation projects or a determined project and the cost of electricity from the grid, highlighting the fact that it is an intuitively clear and simple metric (Loewen, 2020).

$$LCOE = \frac{\sum_{t=0}^T (I_t + O_t + M_t + F_t) / (1 + r)^t}{\sum_{t=0}^T E_t / (1 + r)^t} \quad (1)$$

The LCOE was evaluated in a 25-year period, using a yearly discount rate of 8.08 %, as it is the mean rate for projects in Latin America (Marín & Villada, 2020). As this work attempts to evaluate the reality of renewable energy sources in Colombia. Thus, it includes the tax incentives offered by Law 1715 of 2015 and the cost structure proposed by Castillo-Ramírez et al. (2015) was used to estimate the investment costs (CAPEX) and O&M expenditures (OPEX) for developing each one of the projects. The cost structure is shown in Fig. 3.

Castillo, Mejía & Giraldo developed a tool for estimating the LCOE of different generation technologies for the entire territory of Colombia.

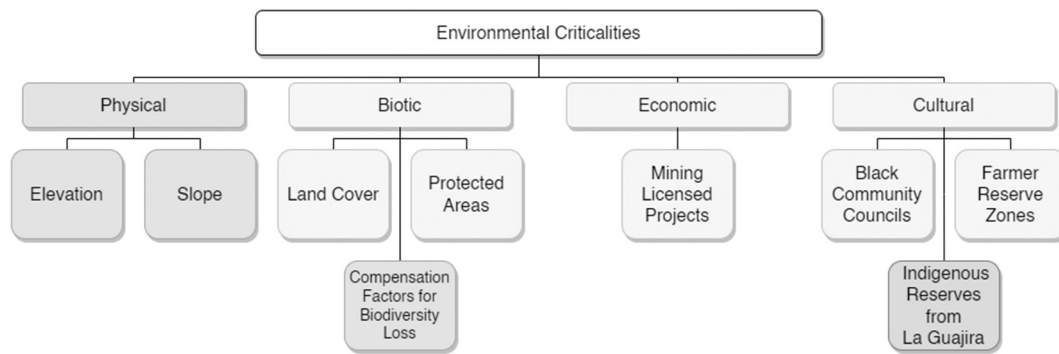


Fig. 2. Environmental criticalities.

Table 2
Experts' sector and field of expertise.

Expert	Sector	Expertise
1	Academic	Physical
2	Private – Energy	Physical-Biotic
3	Private – Energy	Physical
4	Public	Biotic
5	Public	Biotic
6	Private – Environmental	Biotic-Economic-Cultural-Political
7	Academic	Economic-Cultural-Political
8	Public	Economic-Cultural-Political

The costs were updated to the present through the cost reduction curves experienced and projected by International Renewable Energy - IRENA for the South American region (IEA, 2019; IRENA, 2020) and in accordance with suggested new trends inside the renewables business

(Sargent and Lundy, 2020). For the present research, the costs of transmission lines, acquisition or leasing of properties and cost of roads were removed since they will be assumed as a spatial variable within the model that evaluates the complete cost of a project at a certain point. Under these considerations, for solar projects the CAPEX ranged between 0,86 to 1,1 MUSD/MW, and between 1,57 to 1,88 MUSD/MW for wind projects, depending on the specific conditions of each location.

It is worth noting that the criteria to define areas where the resource is not sufficient to develop any of the addressed projects is determined by the LCOE. A non-profitable location is defined when the LCOE is higher than 100 USD/MWh to be competitive in the short term according to recent auctions carried out in Colombia (UPME, 2019). Global averages are around 50 USD/MWh (IRENA, 2020). The LCOE chosen as the limit discards projects that will not be profitable but allows the results to include projects that will become competitive in the short-term.

Table 3
Preference structure results – environmental criticalness.

			Expert								Mean	Std. Deviation
			1	2	3	4	5	6	7	8		
Biotic dimension	Land cover	Xerophytia	6	1	9	9	10	7	10	7	7.4	3.0
		Subxerophytia	5	1	8	9	9	7	8	7	6.8	2.7
		Savannah	3	1	1	7	9	4	6	4	4.4	2.8
		Agroecosystem	2	3	3	9	6	6	5	5	4.9	2.2
		Secondary vegetation	4	4	3	5	7	7	6	5	5.1	1.5
		Desert	2	1	2	9	8	6	9	5	5.3	3.3
		Bushland	4	4	2	8	4	6	6	5	4.9	1.8
		Fragmented forest	5	5	6	10	6	8	9	7	7.0	1.9
		Forest	7	7	10	10	9	8	9	9	8.6	1.2
		Rock complex	3	8	2	8	3	4	6	5	4.9	2.3
		Transformed. Transitional	3	4	2	6	4	7	5	4	4.4	1.6
		Grassland	3	3	2	7	8	5	6	5	4.9	2.1
	Protected areas	Regional integrated management districts	4	6	8	7	9	7	10	7	7.3	1.8
		National integrated management districts	6	8	8	6	9	7	10	8	7.8	1.4
		Soil conservation districts	6	5	4	9	10	8	9	7	7.3	2.1
	Weights	Land cover	25	20	10	25	60	20	40	30	28.8	15.3
		Protected areas	25	50	50	25	20	30	50	35	35.6	12.7
		Compensation factors for biodiversity loss	50	30	40	50	20	50	10	35	35.6	15.0
Cultural dim.	Farmer reserve zones	7	5	9	5	7	9	4	6	6.5	1.9	
	Black community councils	9	7	8	8	7	9	8	8	8.0	0.8	
Economic dim.	Licensed mining	6	5	3	4	10	6	1	3	4.8	2.7	

Table 4
Relative weights of the environmental dimensions.

Dimension	Expert								Resulting weight	Std. Deviation
	1	2	3	4	5	6	7	8		
Physical	20	20	25	35	15	10	40	30	24.4	10.2
Biotic	40	35	25	35	30	20	20	20	28.1	8.0
Economic	20	15	25	10	25	30	20	20	20.6	6.2
Cultural	20	30	25	20	30	40	20	30	26.9	7.0

Step 7: gross potential determination

Locations that fulfill the double condition of being unrestricted and offering an LCOE lower than the threshold established in Step 6 are considered suitable for developing solar PV farms or wind farms. The resulting area after accounting for all these pixels is the total exploitable area. Each renewable energy source has its own power density per area for utility scales. Therefore, the gross potential by technology for continental Colombia can be determined.

Step 8: site suitability maps for wind and solar PV farms

The methodology aims to identify the areas that jointly offer the best financial and environmental conditions. The most attractive projects for developers are expected to be the most profitable and those that require the least environmental management. Hence, environmental criticalities were classified into 3 groups: low, medium, high. The LCOE was also classified to facilitate building the matrix shown in Fig. 4. The highest suitability is associated with scores of 1, and the poorest is associated with scores of 9.

Data sources

The consulted sources are described in Table 5. Most local data is now available on websites corresponding to open data government

programs, such as datos.gov.co (Gómez Colorado et al., 2018) and siac.gov.co (Minambiente, 2023).

Results

The results are displayed in seven different maps, each of them showing the outcome of several calculations performed to achieve the objectives and steps described in the methodology. Fig. 5 shows the resulting maps for solar PV farms, while Fig. 6 does so for wind farms. Resource, Environmental Restrictions, Environmental Criticalities, Capacity Factor, Cost of Investment, LCOE and Financial & Environmental Integration were chosen based on the spatial nature of the variables they represent.

The data in Figs. 5 and 6 was integrated to build the refined solar and wind potential maps for Colombia (Figs. 7 and 8, respectively). To evaluate the use of this maps as a guiding tool for developers, these figures present an overlay of large-scale solar and wind projects (>20 MW in capacity) registered in the Colombian Energy and Mines Planning Unit - UPME. The blue dots represent the projects that have reached feasibility and detailed engineering stages (Phases II and III). The results show that most of the projects in more advanced stage are effectively located in the areas with the best environmental and financial performances identified in the wind and solar maps.

The maps show that there are multiple areas with the necessary characteristics for the easy development and appropriate performance

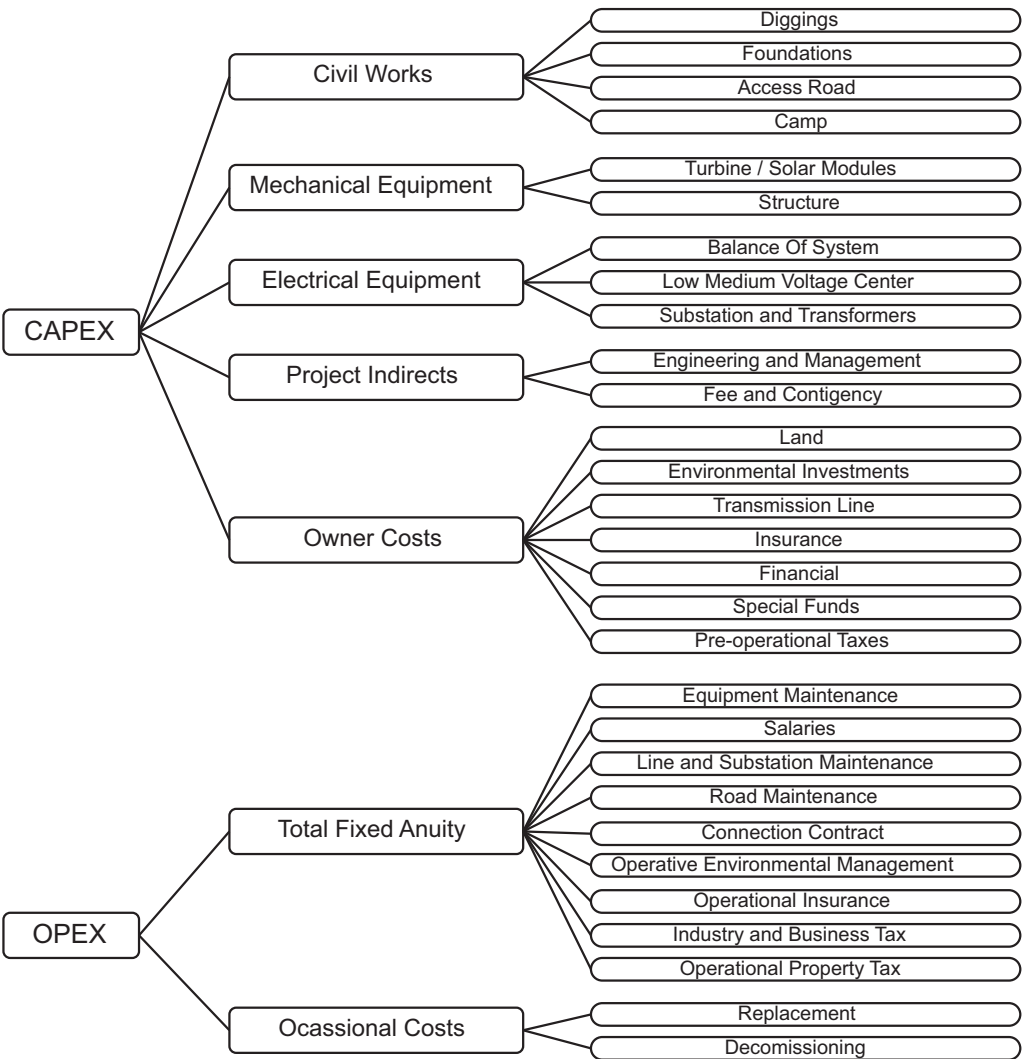


Fig. 3. Cost structure.

of photovoltaic solar farms throughout the Colombian territory. On the north coast of the country, the departments of La Guajira, Córdoba, Sucre, Bolívar, Atlántico, Magdalena and Cesar stand out, characterized by an abundant and persistent resource complemented by favorable environmental conditions. In addition, towards the center of the country are the inter-Andean valleys, highlighting the portion of the Cauca River Valley between the departments of Valle del Cauca and Quindío, and the Magdalena River Valley from the department of Huila until it reaches the Caribbean region. The eastern plains, mainly the departments Meta and Casanare, also have ideal conditions for harnessing the sun's energy on commercial scales, connected to the national grid. The sum of all feasible projects is an installed capacity of over 8 TW, capable of generating >11,000 TWh/year.

Wind is concentrated in the upper part of the Caribbean region, highlighting the municipalities of Uribia, Maicao and Manaure in the department of La Guajira, and Puerto Colombia, Barranquilla, Tubará and Galapa in the department of Atlántico, located on the north coast. In addition to the lower presence of the resource in the country, wind projects are highly restricted by environmental conditions. The established safety distance from populated centers, roads and rivers significantly reduces the area available for harnessing the wind. If all possible wind parks were deployed, 35 GW of installed capacity could be added to Colombia's energy matrix, and 176 TWh could be generated in a year by just exploiting this non-conventional energy sources.

Discussion

This work uses information from different sources. Heterogeneity can be found in the resolution and update of the data, thus inducing some errors when layers were overlapped. Expert ratings incorporate subjectivity to the results, but gathering different academic educations, economic sectors and specific knowledge helps reduce the bias of the obtained results. The processes involved implementing factors, constants and trends that may also add errors to the output of the work. Nevertheless, constant and continuous validation processes, such as comparing the obtained LCOE to global and regional trends, Capacity Factors with research results and, above all, the superposition of real projects in advanced implementation or licensing process stages with the final results, shows that the work is useful as a tool for guiding stakeholders. It provides early warnings of the appearance of commercial projects in their zones, allowing them to be prepared in terms of regulation and local control capacity, especially for solar PV farms.

Colombia has a complex political history, since it has been involved in an armed conflict for over fifty years, and peace processes with some of the armed groups are just now being implemented. This condition shapes the environment into a complex and unique environment. Furthermore, its cultural and biotic wealth imposes special conditions to fully comply with regulations and laws. Moreover, the information to fully reflect this on a map is often fragmentary and incomplete.

This work strived to implement the highest resolution data available that matched the technical conditions required to fit the proposed methodology. A spatial scale of one decimal degree, such as the one

Table 5
Information sources and format.

Layer	Source	Format
Roads	ANI-IGAC (Agencia Nacional de Infraestructura, 2023)	Shapefile
Mining projects	ANLA (Minambiente, 2023)	Shapefile
Ethnic communities	ANT (ANT, 2023)	Shapefile
Peasant reserve zones	ANT (ANT, 2023)	Shapefile
Black community councils	ANT (ANT, 2023)	Shapefile
Wind speed	ECMWF (ECMWF, 2023)	NetCDF
Electrical substations	UPME (UPME, 2023)	PDF-Shapefile
ETCR	ARN (ARN, 2023)	Shapefile
Protected archeological area	ICANH (ICANH, 2023)	Shapefile
Base cartography	IGAC (IGAC, 2023)	Shapefile
Land cover	IDEAM (Minambiente, 2023)	Shapefile
Ecosystems	IAvH-MADS (Minambiente, 2023)	Shapefile
Flood zones	IDEAM (Minambiente, 2023)	Shapefile
Forests	IDEAM-SINCHI (Minambiente, 2023)	Shapefile
SINAP	PNN (Minambiente, 2023)	Shapefile
FCCPB	ANLA (Minambiente, 2023)	Shapefile
Coca crops	SIMCI-UNODC (UNODC, 2023)	Shapefile
Solar radiation (GHI)	Universidad EIA (González-Duque et al., 2018)	Raster
Transmission lines	UPME (UPME, 2023)	PDF-Shapefile
Land cost	UPRA (UPRA, 2023)	Shapefile
DEM	USGS (U.S. Geological Survey, 2019)	Raster

implemented in this study for solar and wind resources, might be easily improved because more precise and better data is constantly being released. The changing nature of the information will be the main reason leading to the need to update these maps and results over time.

It is worth noticing that the scale and information used for this research is limited to analyze utility-scale projects with an installed capacity over 100 MW with commercial technologies available in 2022. This means that the results do not reflect the conditions for development of smaller grid-connected or distributed generation projects, nor for disruptive wind and solar generation technologies with different environmental impacts. Moreover, this research fails to consider the possible opposition or support that energy projects may face from civil society, advocacy groups or grassroots movements.

Considering these limitations, the scope of this work is aimed at facilitating planning strategies in which local, regional, energy and environmental stakeholders enhance their knowledge related to how the characteristics that may affect the implementation of wind farms and solar parks are distributed throughout the territory. This work never attempted to replace the process of on-the-ground information gathering required to apply to environmental licenses, nor the detailed financial and technical assessments needed for a particular energy project.

Conclusions

The results of this work are a tool to facilitate the initial search for prospective zones for solar and wind farms for private investors, stakeholders and government authorities from the energy, environmental and social sectors. Municipal and departmental officials will also benefit from understanding the potential in their region and the possibility of directing policies towards using it.

The financial analysis was focused on the levelized cost of electricity (LCOE), a metric used to analyze the costs and performance of generation projects based on non-conventional renewable sources. This facilitated the comparison with global and regional trends, and between technologies. It also allowed the results to be analyzed in the context of recent renewable energy auctions carried out in Colombia.

The methodology proposed here could be applied in other countries or territories. However, since the environment is comprised of physical,

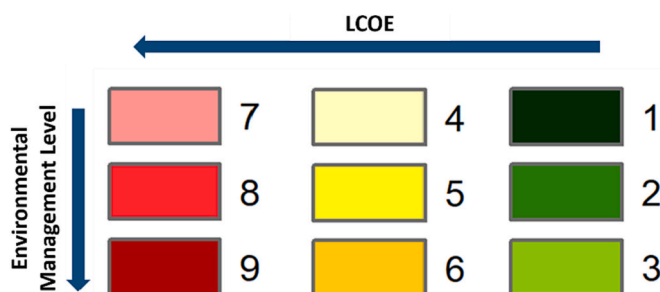


Fig. 4. Environmental and financial integration matrix.

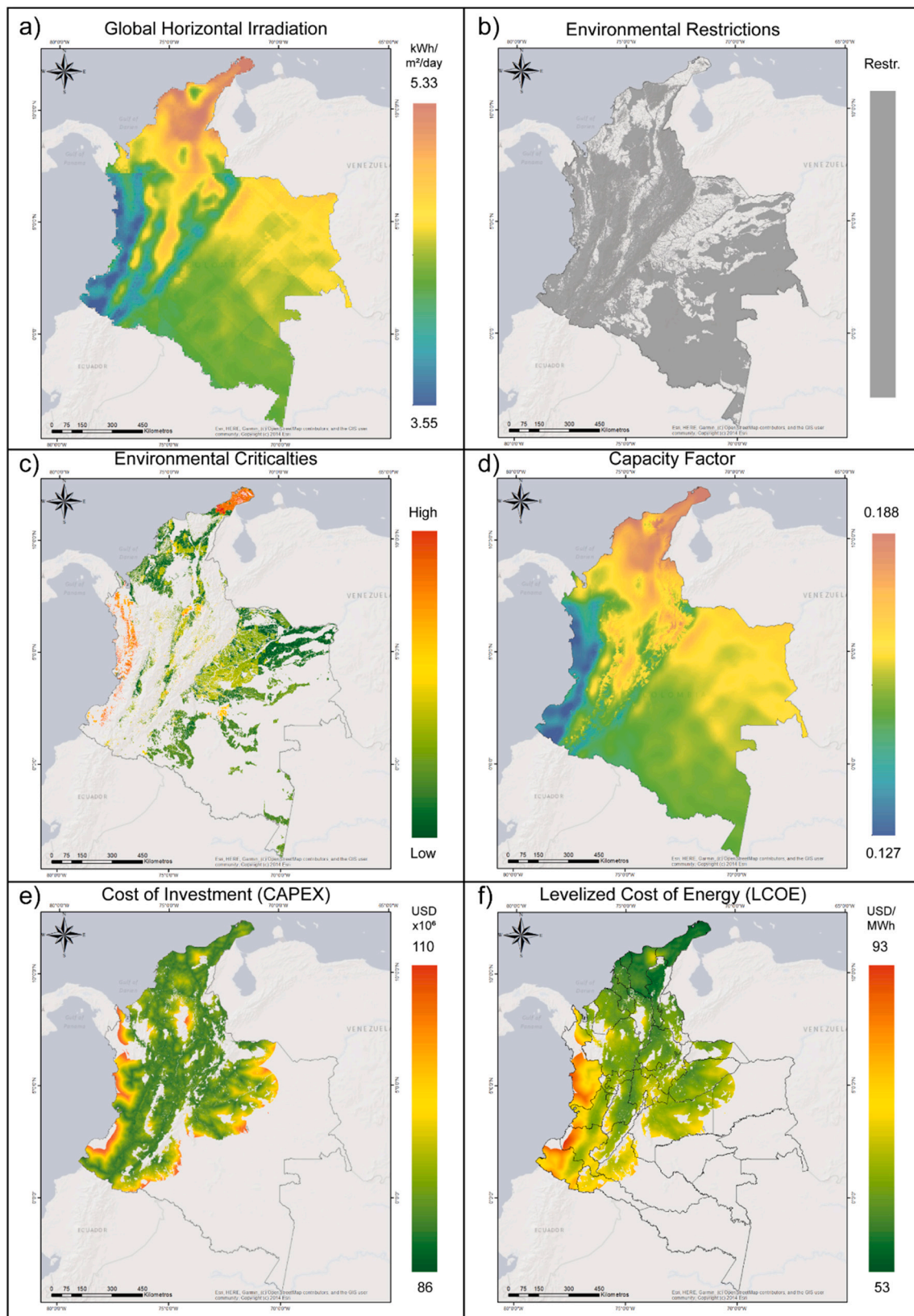


Fig. 5. Resulting maps for solar PV farms. a) Global Horizontal Radiation in kWh/m²/day b) Areas with environmental restrictions for solar power, shown in a darker shade c) Level of environmental criticalities d) Capacity factor of solar PV farms e) Cost of investment in million US dollars for a 100 MW plant f) Levelized cost of energy in USD/MWh.

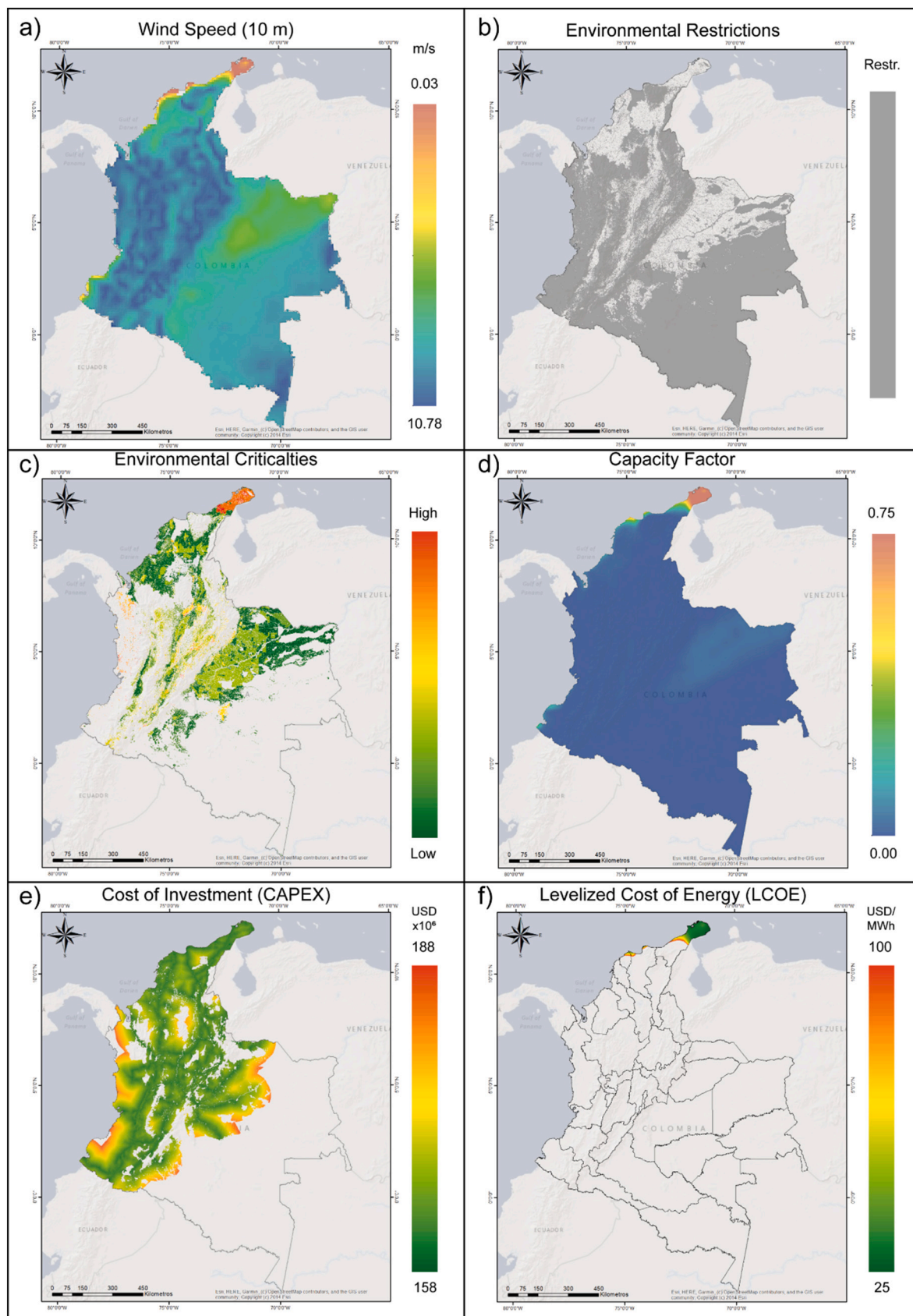


Fig. 6. Resulting maps for wind farms. a) Wind speed at a 10 m altitude in m/s b) Areas with environmental restrictions for wind energy, shown in a darker shade c) Level of environmental criticalities for wind energy d) Capacity factor of wind farms e) Cost of investment in million US dollars for a 100 MW plant f) Levelized cost of energy in USD/MWh.

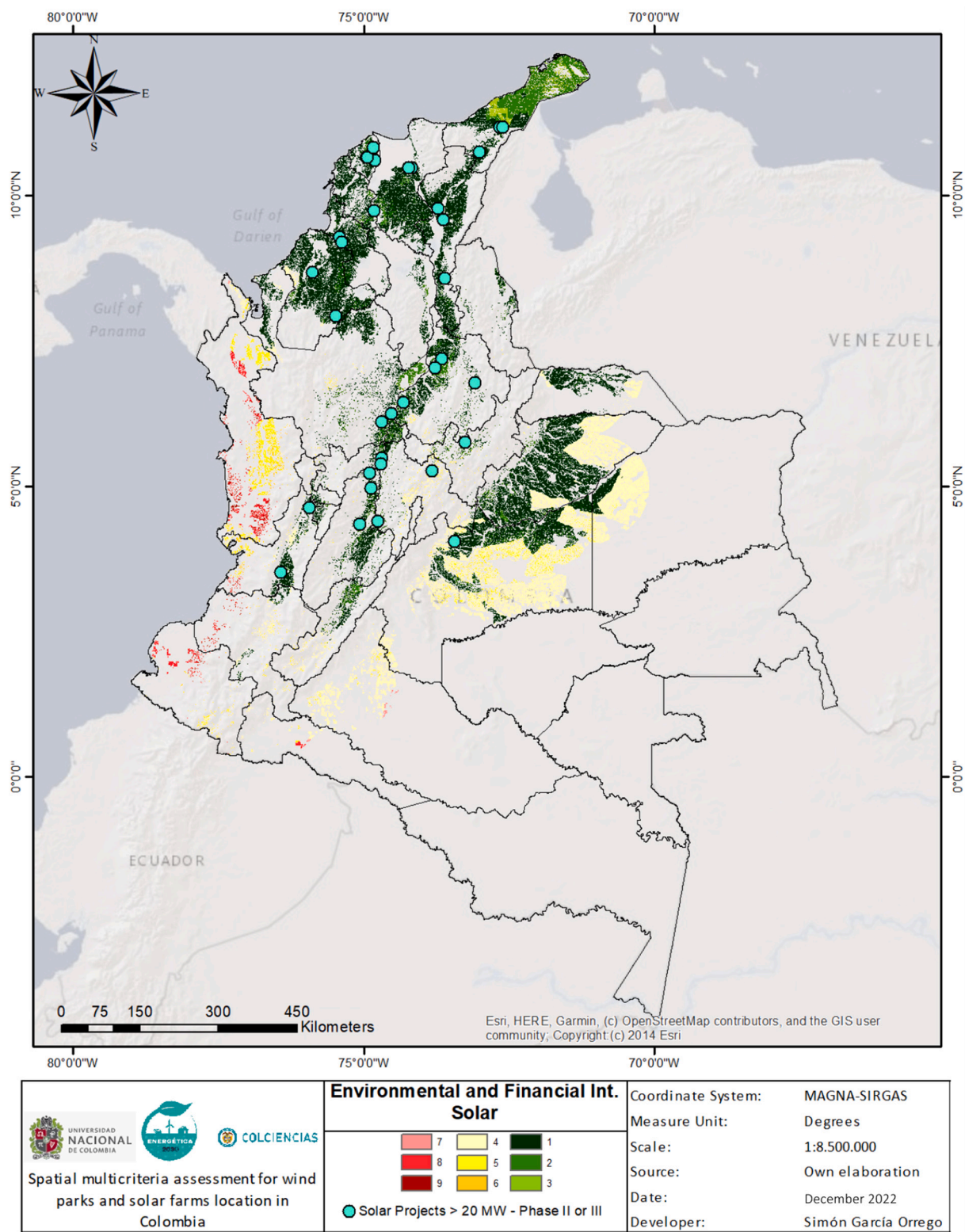


Fig. 7. Resulting map for the integration of the financial and environmental performance of solar PV farms. Projects under development with a capacity of 20 MW or higher are shown overlaid.

biotic, economic, cultural and political characteristics, each country will configure its own set of restrictions and criticalities. Therefore, applying this methodology in other contexts will require specific and information derived from a local analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

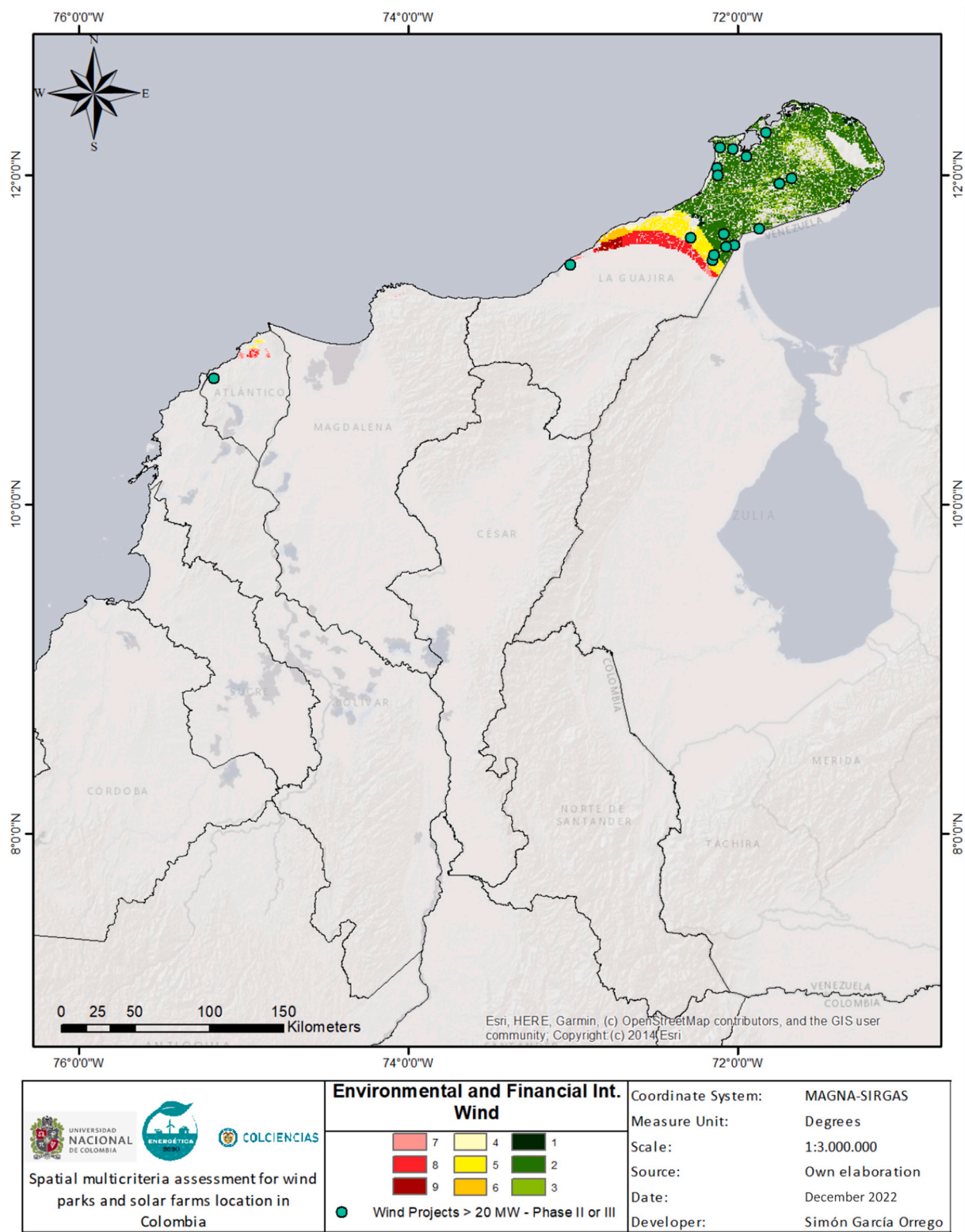


Fig. 8. Resulting map for the integration of the financial and environmental performance of wind parks. Projects under development with a capacity of 20 MW or higher are shown overlaid.

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